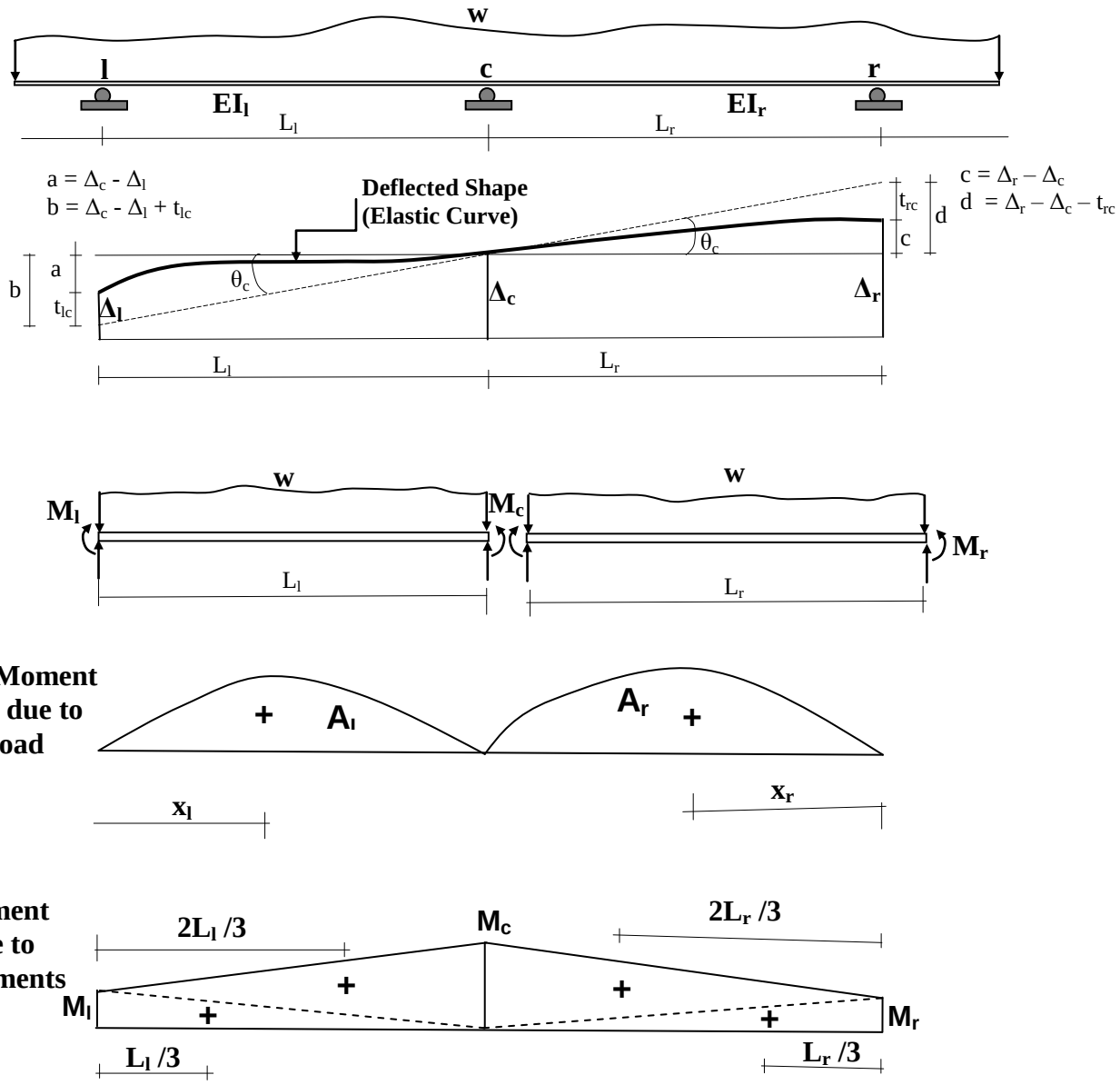
Three Moment Equation

The three-moment equation is an expression that relates the moments and displacements at three points (preferably at supports) in a continuous member such as continuous beams. The relation can be derived by considering part of a continuous beam subjected to arbitrary loading as shown below:



For a continuous member, the condition of compatibility must be satisfied at each point. For example the slope at point c (middle support) is the same whether it is computed by using the left or right segment of the beam.

Thus: $$\frac{\Delta_c - \Delta_l + t_{lc}}{L_l} = \frac{\Delta_r - \Delta_c - t_{rc}}{L_r}$$

Rearranging the equations:

$$\frac{t_{lc}}{L_l} + \frac{t_{rc}}{L_r} = \frac{\Delta_l - \Delta_c}{L_l} + \frac{\Delta_r - \Delta_c}{L_r}$$

From the second theorem of moment – area method:

$$t_{lc} = \frac{1}{EI_l}\left[A_l\, x_l + \frac{M_l\, L_l}{2}\left(\frac{L_l}{3}\right) + \frac{M_c\, L_l}{2}\left(\frac{2L_l}{3}\right)\right]$$

$$t_{rc} = \frac{1}{EI_r}\left[A_r\, x_r + \frac{M_r\, L_r}{2}\left(\frac{L_r}{3}\right) + \frac{M_c\, L_r}{2}\left(2\frac{L_r}{3}\right)\right]$$

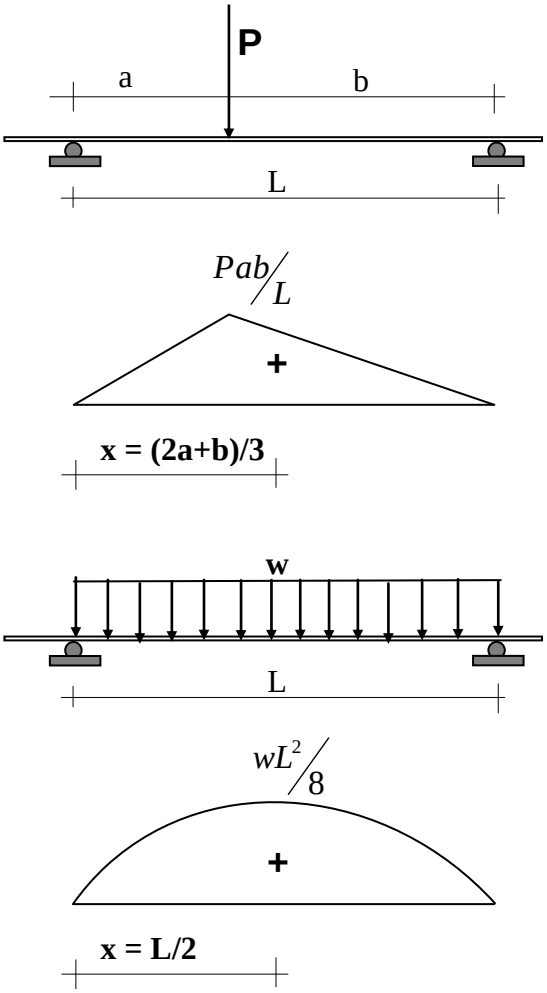
Inserting these equations and rearranging:

$$M_l\left(\frac{L}{EI}\right)_l + 2Mc\left(\frac{L_l}{EI_l} + \frac{Lr}{EIr}\right) + Mr\left(\frac{L}{EI}\right)_r = -6\left(\frac{Ax}{EIL}\right)_l - 6\left(\frac{Ax}{EIL}\right)_r + 6\left(\frac{\Delta_l - \Delta_c}{L_l} + \frac{\Delta_r - \Delta_c}{L_r}\right)$$

If $EI_l = EI_r$ and there is no support settlements, the above equation reduces to:

$$M_l\, L_l + 2M_c\,(L_l + L_r) + M_r\, L_r = \sum -6\left(\frac{Ax}{L}\right)$$

The expression (6Ax) /L in the case of concentrated and uniform load are found to be:



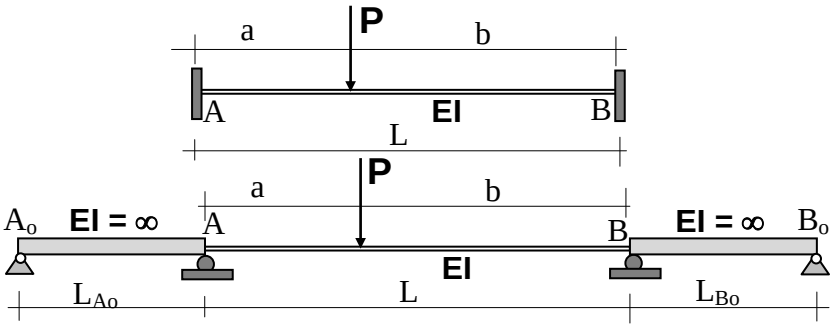
$$\frac{6Ax}{L} = \frac{Pa(L^2 - a^2)}{L}$$

$$\frac{6Ax}{L} = \frac{wL^3}{4}$$

In the special case where the continuous beam has a fixed support at one /both ends, an outer imaginary equivalent beam of finite span with infinite stiffness is considered to change the system to an equivalent continuous beam. The fixed end moments for the case of ether both ends fixed or one end fixed and the other end hinged can be determined with this approach. The fixed end moments due to the effect of concentrated and uniformly distributed load for the case of both ends fixed is presented in the following examples to demonstrate how the three moment equation is very much suitable and powerful means to determine fixed end moments for different case of loading.

Example 1

Determine the end moment for the beam under the effect of concentrated load P as shown below



The three moment equation applied to spa A_oA and AB is written as:

$$0 + 2M_A\left(\frac{L_{Ao}}{\infty} + \frac{L}{EI}\right) + M_B\left(\frac{L}{EI}\right) = -\frac{Pb(L^2 - b^2)}{L\,EI} \rightarrow 2M_A + M_B = -\frac{Pb(L^2 - b^2)}{L^2}$$

Similarly the three moment equation for span AB and BB_o is written as:

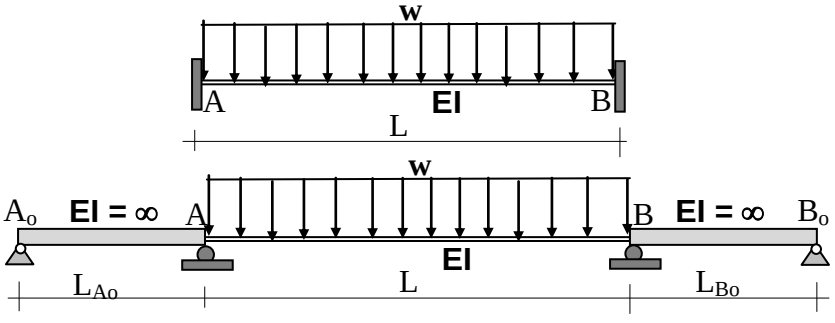
$$M_A\left(\frac{L}{EI}\right) + 2M_B\left(\frac{L}{EI} + \frac{L_{Bo}}{\infty}\right) + 0 = -\frac{Pa(L^2 - a^2)}{L\,EI} \rightarrow M_A + 2M_B = -\frac{Pa(L^2 - a^2)}{L^2}$$

Solving the two equations simultaneously:

$$M_A = -\frac{Pab^2}{L^2} \quad ; \quad M_B = -\frac{Pa^2b}{L^2}$$

Example 2

Determine the end moment for the beam under the effect of uniformly distributed load w as shown below



$$\frac{6Ax}{L} = \frac{wL^3}{4}$$

The three moment equation applied to spans A_oA and AB is written as:

$$0 + 2M_A\left(\frac{L_{Ao}}{\infty} + \frac{L}{EI}\right) + M_B\left(\frac{L}{EI}\right) = -\frac{wL^3}{4EI} \rightarrow 2M_A + M_B = -\frac{wL^2}{4}$$

Similarly the three moment equation for spans AB and BB_o is written as:

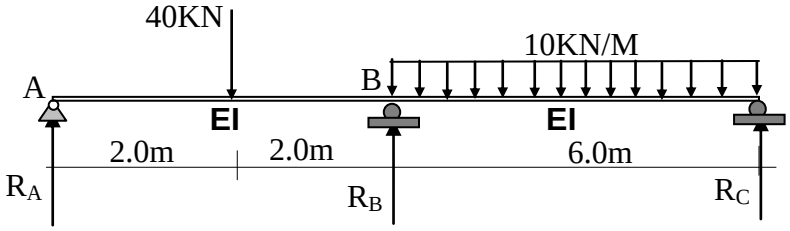
$$M_A\left(\frac{L}{EI}\right) + 2M_B\left(\frac{L}{EI} + \frac{L_{Bo}}{\infty}\right) + 0 = -\frac{wL^3}{4EI} \rightarrow M_A + 2M_B = -\frac{wL^2}{4}$$

Solving the two equations simultaneously:

$$M_A = -\frac{wL^2}{12} \quad ; \quad M_B = -\frac{wL^2}{12}$$

Example 3

Determine the reaction and draw the sheer force and bending moment diagrams for the following beam.



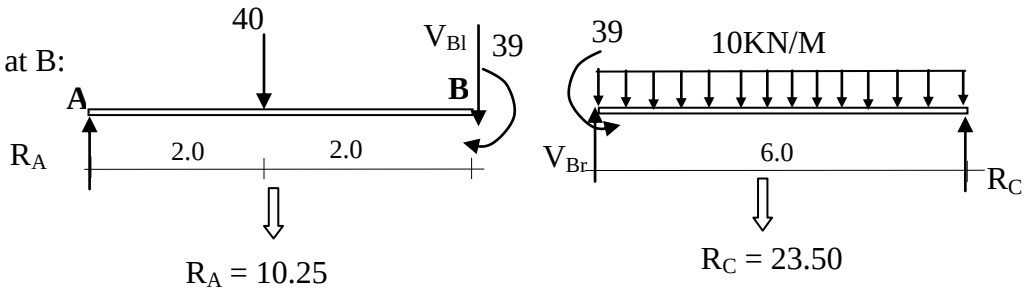
For beams with constant flexural stiffens, the three moment equation is written as

$$M_A L_{AB} + 2M_B\left(L_{AB} + L_{BC}\right) + M_C L_{BC} = \sum -6\left(\frac{Ax}{L}\right)$$

$$0 + 2M_B(4+6) + 0 = -\left[\frac{40*2(4^2 - 2^2)}{4} + \frac{10*6^3}{4}\right] = -780 \rightarrow M_B = -39KNM$$

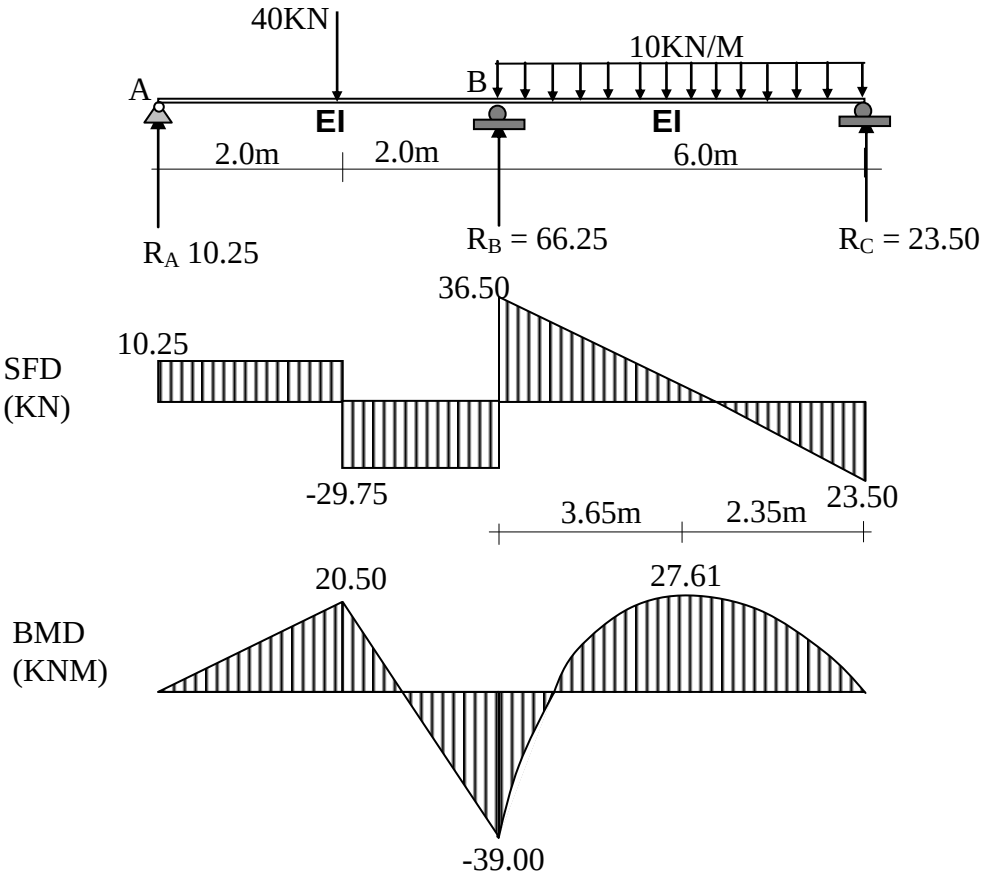
Taking section at B and analyzing the left and right part

Section at B:



The Remaining reaction at the support B can be determined considering equilibrium condition of the entire beam and found to be R_B = 66.25KN

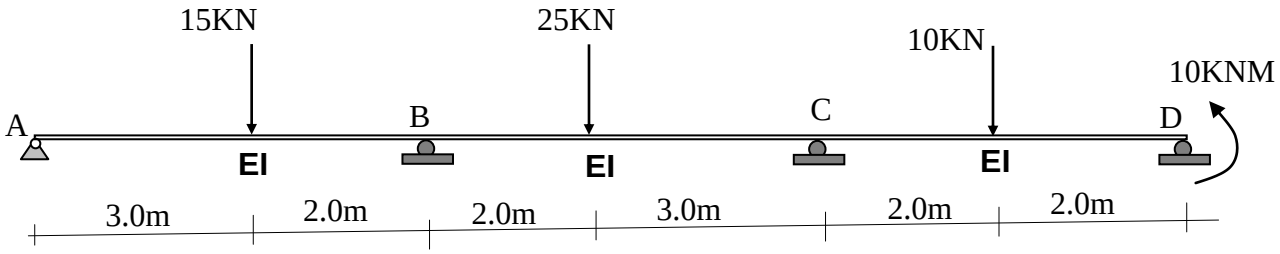
Shear force and bending moment diagrams



Exercise

Using three moment equation, determine the support moments and draw the sheer force and bending moment diagrams for the following beams.

(a)



(b)

